Clemson University

Heat Transfer Project Report

Transient Heat Conduction in a Cylindrical Multi-Layer Wall

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I. Problem Description and Objectives

The main objective of this project is to model transient heat transfer using real-world materials and geometries. The model is analyzed using the finite difference method (FDM) to evaluate the temperature distribution in a cylindrical multi-layered system as functions of both radial distance and time. To accurately verify the analytical system produced in this report, the sensitivity, validation, and stability of the model must be evaluated. This ensures that the analysis produces realistic results that reflect physical behavior. Additionally, the differences between steady-state and transient systems are compared and analyzed. The concepts covered in this report allow students to effectively model heat transfer problems and understand the effects of multi-layered geometries.

II. System Geometry and Material Properties

The system analyzed in this report was modeled after a insulated exhaust pipe. This was chosen due to an exhaust meeting all project requirements and making the project apply to a real-world application. This system consists of a inner pipe carrying a hot exhaust gas (T_{in}), a intermediate insulation to reduce heat transfer in the system and a outer shell to protect from hazards. Each of the materials listed above have their properties listed in Table 1. These properties will dictate the level of heat transfer occurring through each material.

TABLE 1: Material properties used for modeling

Material	Density (ρ) [kg/m^3]	Conductivity (k) [W/m-K]	Heat Capacitance (C_p) [J/kg-K]
Stainless Steel	7900	16	500
Insulation	100	0.1	1000
Aluminum	2700	205	900

The boundary conditions are listed below in Table 2 where T_{in} is the entering exhaust heat and T_∞ is the surrounding air temperature outside of the exhaust pipe. These boundary conditions are given to simplify the problem and are consistent in Dirichlet's numerical method to find transient heat transfer.

TABLE 2: Boundary conditions and geometry constraints

Property	Value
Temperature In [T_{in}] (ºC)	400
Surrounding Temperature [T_∞] (ºC)	25
Initial Temperature (inside) [T_o] (ºC)	25
Inside Exhaust Radius [r_o] (m)	0.050
Stainless Steel Thickness [t_{ss}] (m)	0.040
Insulation Thickness [t_{ins}] (m)	0.003
Aluminum Thickness [t_a] (m)	0.003

III. Numerical Methodology

One-dimensional transient heat transfer in the radial direction with no internal heat generation is shown below in Eq. 1. Where on the left side we see how temperature changes with time and the right explains standard conduction with a cylindrical component to adjust the geometry.

$$\frac{\partial T}{\partial t} = \alpha\left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r}\frac{\partial T}{\partial r}\right) \tag{1}$$

This governing equation uses temperature (T) as a function of both time and direction, α is the thermal diffusivity and r is the radial position. To perform these equations Finite differential Approximations are used which give the Eq. 2-4 below.

$$\frac{\partial T}{\partial t} \approx \frac{T_i^{n+1} - T_i^n}{\Delta t} \tag{2}$$

$$\frac{\partial^2 T}{\partial r^2} \approx \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta r} \tag{3}$$

$$\frac{1}{r}\frac{\partial T}{\partial r} \approx \frac{1}{r_i}\frac{T_{i+1}^n - T_{i-1}^n}{2\Delta r} \tag{4}$$

The equations replace the derivatives with finite differences. Where a time step is represented with n and the radial position is represented by i. Eq. 2-4 will then be substituted into Eq. 1 to solve for the new temperature at a given time and radius which will be used in the analytical model for the system, which is given below in Eq. 5.

$$T_i^{n+1} = T_i^n + \frac{\alpha \Delta t}{(\Delta r)^2}\left[T_{i+1}^n - 2T_i^n + T_{i-1}^n + \frac{\Delta r}{2r_i}(T_{i+1}^n - T_{i-1}^n)\right] \tag{5}$$

The initial conditions of the system are stated in Table 2; however they will be restated as they are important to the simulation. The simulation will begin at t = 0 where inside the exhaust pipe is 25 ºC (room temperature) once the simulation starts the exhaust will begin to run through the pipe at a temperature of 400 ºC. The boundary conditions of the experiment will be set using Dirichlet's ideation which fixes both the boundary conditions of a system to a constant value. This simplifies the analysis to give both the inside and the outside of the pipe a constant start and stop condition.

For the analysis to stay in a stable condition the Fourier's Number must be computed. This number is dependent on the time it takes for heat to spread, how fast it spreads and the spacing between the radial nodes. When this number is less than 0.5 error dies out and becomes a stable analysis. This concept is shown below in Eq. 6 below.

$$0.5 < \frac{(\Delta r)^2}{\Delta t \alpha_{max}} \tag{6}$$

However, when this equation is rearranged, it allows for a stable time step to be computed for our analysis which is shown below.

$$\Delta t < \frac{(\Delta r)^2}{2\alpha_{max}} \tag{7}$$

IV. Implementation

The model was implemented using MATLAB with a script divided into numerous sections including input parameters, material property, material property assignment based on the system geometry, numerical solution and visualization. The cylindrical wall geometry was defined using the inner radius along with the thickness for the insulation, steel and aluminum. A grid was set up using a specified node count and a spatial time step to compute stable node spacing. Material properties were assigned to each node based on the radial position and whether it fell in between the materials bounds. This allowed the model to account for material properties such as changes in thermal conductivity density and specific heat across the geometry.

The transient solution was found using a time-stepping loop. At each step the temperature at a specific radial node was taken for the entire duration of the analysis. Afterward the analysis would move to the next node and repeat the calculation for the temperature. Boundary nodes were updated using the prescribed temperature conditions. The time steps were dictated using the stability conditioned outlined in Eq. 7. Temperature values were stored at each time and radial position in a large matrix this allowed for a full transient behavior graph to be captured. Additional plots were created for stability comparison, sensitivity to spatial distancing, sensitivity to time step, and the validation vs. a steady state solution.

V. Results

The results of the analysis are displayed below. In Fig. 1 the data displays a exponentially decaying graph where the slopes of the graphs slow as the time increases. This shows that as time increases the materiel's temperature will increase until it reaches its steady state temperature profile which is close to t =120s. The slope is fastest at its inner surface however heat transfer slows as it reaches the insulation layer.

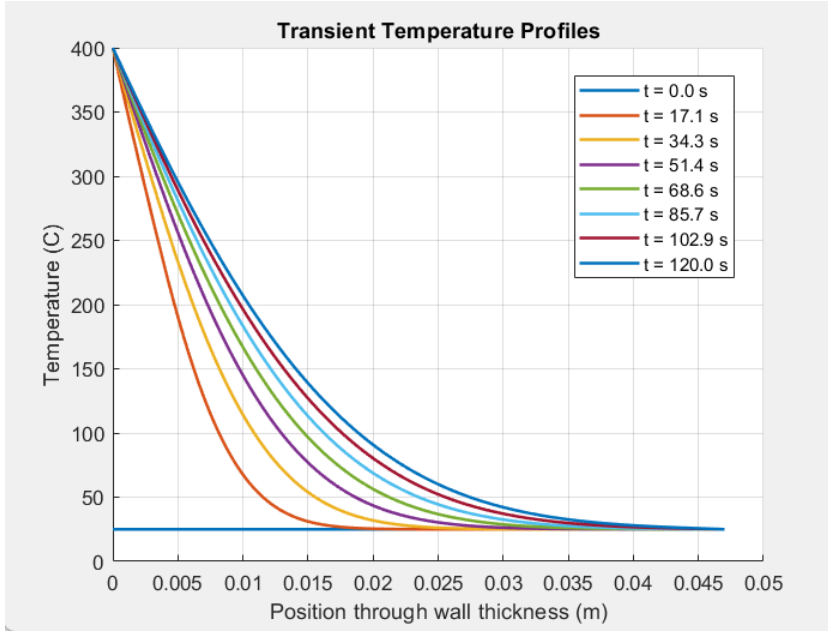

FIGURE 1: This figure displays the transient heat transfer of the cylindrical exhaust system. It correlates the position of the cylinder, the time that has passed and the temperature.

The stability of the analytical model is tested, and the results are shown in Fig. 2-3. In Fig. 2 the stability case has the Fourier's Number less than 0.5 this affects the time step to create a stable case. It shows a quick ten second model of the transient heat transfer where the result is a stable exponential decay that stabilizes at 25 ºC. In Fig. 3 we can see that the unstable case which uses a very high Fournier's Number to clearly relay the systems instability with its given time step. The oscillations become extremely large and unstable over a short period of time.

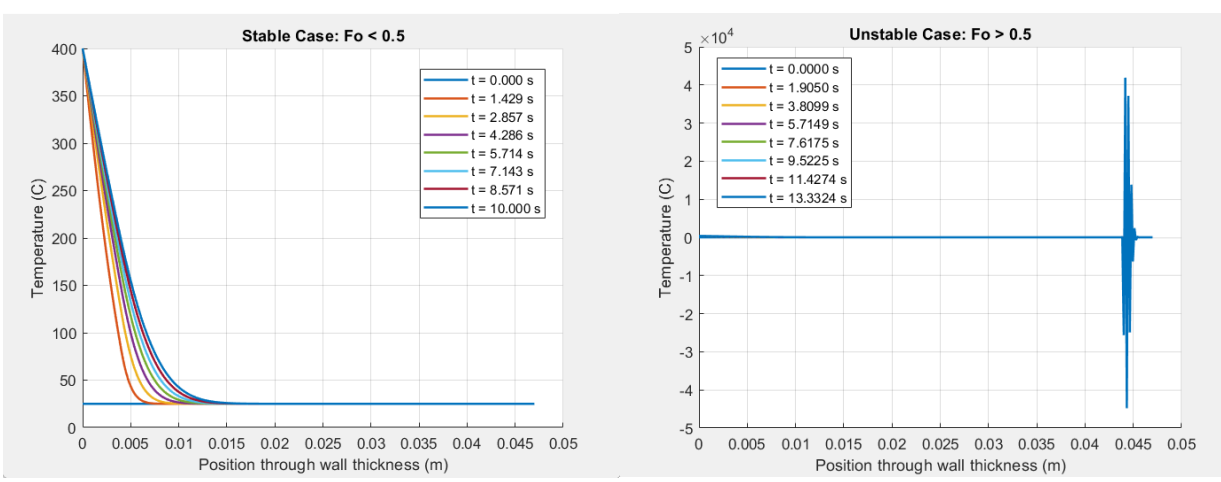

FIGURE 2: This displays both the stable and unstable cases of transient heat transfer through a cylindrical object. It shows two plots with different delta t values based off the Fourier's Number.

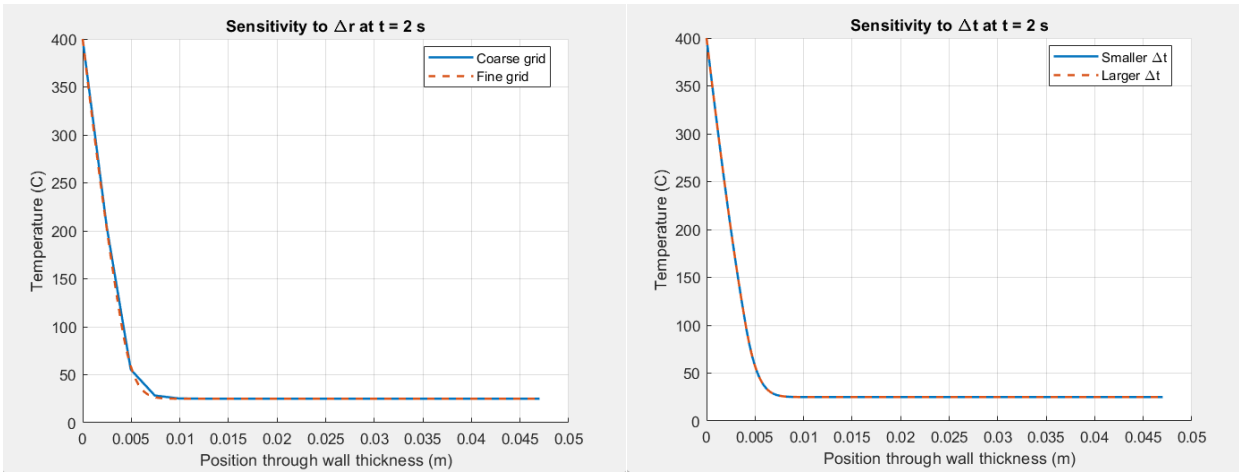

FIGURE 3: This displays the sensitivity of delta t and delta r and a specific point in time.

The sensitivity graphs show the discrepancy between a large amount of time steps vs a smaller amount of time steps. The graph shows that the results of having a large time step vs small time step doesn't matter as both show extremely similar results. For the radial node comparison between a course vs fine grid there is a small variation however this variation is not noticeably enough to cause data to be misconstrued.

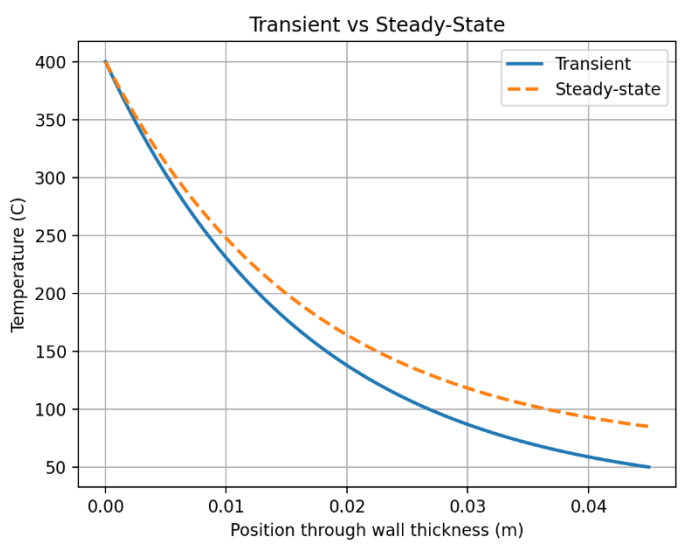

FIGURE 4: This graph displays the transient line produced vs the steady state validation line.

The transient temperature profile gradually approaches the steady-state distribution over time, showing the system's progression toward thermal equilibrium. Differences between the curves indicate

that the wall has not yet fully reached steady state, while the overall shape confirms that insulation dominates the temperature drop due to its low thermal conductivity.

TABLE 3: This table shows the average heat flux for each of the materials used in the analysis

Material	Stainless Steel	Aluminum	Insulation
Heat Flux (q) [W/m^2]	338386	75170	724.21

Table 3 shows the average heat flux for each of the materials. The Aluminum and Stainless Steel both have extremely high conductivity and heat diffusivity therefore they both have high heat flux. The insulation has a comparable lower heat flux due to its properties that slow heat transfer. The aluminum has a much lower heat flux than that of the stainless steel due to the material being toward the end of the transient heat conduction where the temperature is much lower.

VI. Discussion

The results of the transient heat conduction analysis showed the expected physical and numerical behavior of a multi-layered cylindrical wall. The transient temperature profile shows how the geometry heats up and gradually moves radially along the geometry. As time progresses the transient heat transfer will reach its steady state heat transfer profile which indicates that the system has reached thermal equilibrium.

The comparison between the transient and steady-state temperature distributions is used to validate the numerical model. The steady state solution which is derived for cylindrical conduction using a logarithmic equation. The transient solution closely follows the steady state profile however it has slight deviations. If the model did not have a finite time to compute the solution and allowed the transient solution to reach its true steady state position the heat profiles would be almost identical which allows for validation the analysis. The sensitivity analysis was performed to evaluate the effects of the spatial and temporal steps used to solve the solution. The comparison between both the spatial and radial sensitivities are similar therefore the chosen grid resolution has little effect on the outcome. With a sufficient grid even in the insulation layer where the gradient is the steepest. The time step analysis shows that with smaller time steps there is a more accurate result however a larger time step gives a more consistent temperature profile. The time step sensitivity will be negligible if the values satisfy the stability criterion.

The stability analysis highlighted the importance of the Fourier's Number in finite difference analysis. For the time steps that satisfy the criterion there is a smooth exponential decay to the boundary condition which is physically realistic. However, when the criterion is not obeyed the solution exhibits a non-realistic oscillation at the boundary condition where a rapid growth of numerical errors occurs. The results comparison of stability limits the finite difference method and demonstrates the errors that can be found when stability criterion is not met.

Overall, the model was successful in modeling a transient and steady state behavior of a 3 layered cylindrical wall. The validation combined with the analysis on sensitivity conditions and stability demonstrates the validity of the model.

VII. Conclusion

The transient heat conduction analysis of a multi-layered cylindrical wall was profoundly demonstrated through the gradual conduction of heat from its initial temperature profile to its steady state. The results showed that heat transfer is strongly influenced by material properties and wall thickness, which was demonstrated through graphs and through the heat flux through each material. The numerical model was validated through a analytical comparison of the transient heat profile to its steady state counterpart. The sensitivity analysis indicated that the solution was not grid size dependent if the stability criterion was met. The stability analysis demonstrated that the Fourier number was pivotal to avoid nonphysical behavior in the system and visually represented a stable and unstable outcomes. Overall the explicit finite difference method proved a reliable analytical solution to model transient heat conduction when proper stability criterion is used.

VIII. References

Usman, M. (2026). *Explicit finite difference discretization of 1D transient heat conduction in a cylinder*. Clemson University, 2026